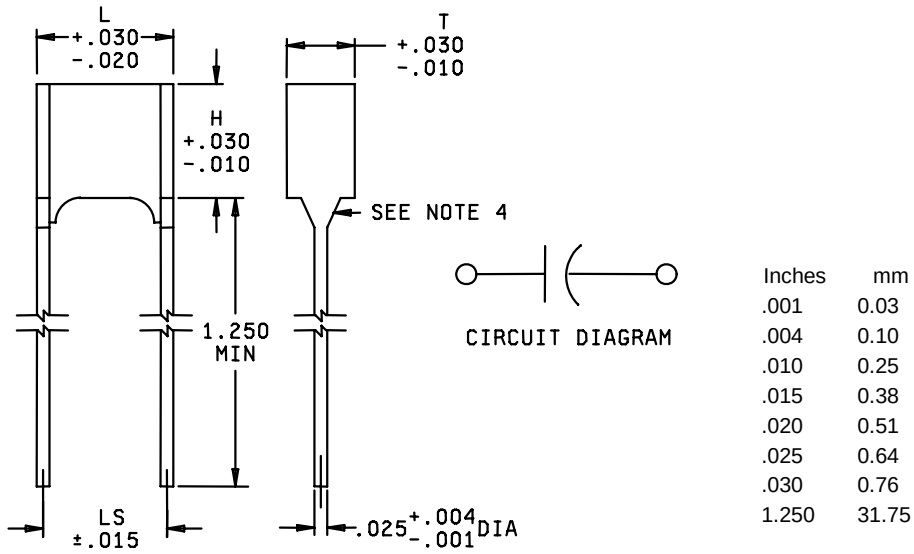
INCH-POUND
MIL-PRF-39022/11D
6 October 2008
SUPERSEDING
MIL-PRF-39022/11C
24 May 1999

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DIRECT-CURRENT, HERMETICALLY SEALED, STYLES CHR20, CHR21, CHR22, CHR23, CHR24, CHR25

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39022.



Inches	mm
.001	0.03
.004	0.10
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
1.250	31.75

Styles CHR20, CHR21, CHR22, CHR23, CHR24, CHR25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. The maximum solder meniscus on the leads shall be .050 (1.27 mm).
5. Lead length may be a minimum of 1 inch long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Capacitor configuration.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case: Hermetically sealed.

Capacitance value: See table I.

Capacitance tolerance: F (1 percent), G (2 percent), J (5 percent), and K (10 percent).

Rated voltage: See table I for +85°C ratings. For +125°C operation, linearly derate +85°C rating by 50 percent.

Operating temperature range: -55°C to +125°C.

Failure rate level (FRL): M (1.0 percent), P (0.1 percent), R (0.01 percent), and S (0.001 percent).

Preconditioning: Not applicable.

Flash point of impregnant: Not applicable.

Thermal shock: Method 107 of MIL-STD-202, condition B.

Radiographic inspection (x ray): Not applicable.

Burn-in: DC: 48 hours at 125°C at 1.4 times the derated 125°C dc voltage rating. The current to each capacitor shall be limited to 1 ampere maximum by a series resistor.

AC: Not applicable.

Seal: Method 112 of MIL-STD-202, test condition C, procedure IIIa, with a leakage rate sensitivity of 10^{-6} atm cm^3/s. After test condition C is completed, test condition A or test condition D shall be performed.

Dielectric withstanding voltage (DWV): In accordance with MIL-PRF-39022, except the charge time shall be 60 seconds minimum, 120 seconds maximum.

Insulation resistance (IR): In accordance with MIL-PRF-39022. When measured at the applicable test temperature and rated voltage, the insulation resistance shall equal or exceed the following values:

Terminal to terminal

Temperature	+25°C	+85°C	+125°C
Megohm x Microfarads	100,000	7,000	700
Except the insulation resistance in megohms need not exceed:	200,000	70,000	7,000

Terminal to case: Not applicable.

Capacitance: Method 305 of MIL-STD-202.

Test frequency: 1 kHz ±100 Hz.

Limit of accuracy: Within ±0.05 percent.

Dissipation factor (DF): Shall not exceed .2 percent at 1 kHz. Limit of accuracy shall be within ±0.05 percent.

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TABLE I. Capacitor characteristics and dimensions.

Part or Identifying Number (PIN) 1/	DC rated voltage	Capacitance Value in µF	Capacitance tolerance	Dimensions 2/			
				T 3/	H 3/	L 3/	LS
CHR20							
M39022/11A333--	30	.033	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11A393--	30	.039	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11A473--	30	.047	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11A563--	30	.056	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11A683--	30	.068	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11A823--	30	.082	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11A184--	30	.18	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11A224--	30	.22	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11A274--	30	.27	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11A334--	30	.33	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11A394--	30	.39	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11A474--	30	.47	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11A564--	30	.56	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11A684--	30	.68	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11A824--	30	.82	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11A105--	30	1.0	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
CHR21							
M39022/11B103--	50	.01	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11B123--	50	.012	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11B153--	50	.015	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11B183--	50	.018	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11B223--	50	.022	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11B273--	50	.027	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11B333--	50	.033	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11B393--	50	.039	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11B473--	50	.047	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11B683--	50	.068	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11B823--	50	.082	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11B104--	50	.10	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11B124--	50	.12	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11B154--	50	.15	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11B184--	50	.18	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11B224--	50	.22	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11B274--	50	.27	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11B334--	50	.33	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11B394--	50	.39	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11B474--	50	.47	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11B564--	50	.56	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11B684--	50	.68	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
CHR22							
M39022/11C562--	100	.0056	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11C682--	100	.0068	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11C822--	100	.0082	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11C103--	100	.01	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11C123--	100	.012	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11C153--	100	.015	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11C333--	100	.033	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11C393--	100	.039	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11C473--	100	.047	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11C563--	100	.056	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11C683--	100	.068	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11C823--	100	.082	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)

See footnotes at end of table.

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TABLE I. Capacitor characteristics and dimensions. – Continued.

Part or Identifying Number (PIN) 1/	DC rated voltage	Capacitance Value in µF	Capacitance tolerance	Dimensions 2/			
				T 3/	H 3/	L 3/	LS
CHR22							
M39022/11C104--	100	.10	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11C124--	100	.12	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11C154--	100	.15	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11C184--	100	.18	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11C224--	100	.22	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
CHR23							
M39022/11D332--	150	.0033	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11D392--	150	.0039	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11D472--	150	.0047	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11D562--	150	.0056	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11D682--	150	.0068	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11D223--	150	.022	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11D273--	150	.027	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11D333--	150	.033	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11D393--	150	.039	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11D563--	150	.056	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11D683--	150	.068	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11D823--	150	.082	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11D104--	150	.10	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
CHR24							
M39022/11E222--	200	.0022	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11E272--	200	.0027	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11E332--	200	.0033	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11E392--	200	.0039	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11E472--	200	.0047	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11E123--	200	.012	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11E153--	200	.015	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11E183--	200	.018	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11E223--	200	.022	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11E273--	200	.027	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11E333--	200	.033	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11E393--	200	.039	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11E473--	200	.047	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11E563--	200	.056	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11E683--	200	.068	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
CHR25							
M39022/11F102--	300	.001	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11F122--	300	.0012	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11F152--	300	.0015	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11F182--	300	.0018	F, G, J, K	.110 (2.79)	.210 (5.33)	.240 (6.10)	.200 (5.08)
M39022/11F222--	300	.0022	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11F272--	300	.0027	F, G, J, K	.170 (4.32)	.270 (6.86)	.240 (6.10)	.200 (5.08)
M39022/11F332--	300	.0033	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F392--	300	.0039	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F472--	300	.0047	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F562--	300	.0056	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F682--	300	.0068	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F822--	300	.0082	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F103--	300	.01	F, G, J, K	.170 (4.32)	.270 (6.86)	.340 (8.64)	.300 (7.62)
M39022/11F123--	300	.012	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11F153--	300	.015	F, G, J, K	.190 (4.83)	.380 (9.65)	.340 (8.64)	.300 (7.62)
M39022/11F183--	300	.018	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)

See footnotes at end of table.

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TABLE I. Capacitor characteristics and dimensions. – Continued.

Part or Identifying Number (PIN) 1/	DC rated voltage	Capacitance Value in µF	Capacitance tolerance	Dimensions 2/			
				T 3/	B 3/	L 3/	LS
CHR25							
M39022/11F223--	300	.022	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11F273--	300	.027	F, G, J, K	.190 (4.83)	.380 (9.65)	.450 (11.43)	.400 (10.16)
M39022/11F333--	300	.033	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)
M39022/11F393--	300	.039	F, G, J, K	.220 (5.59)	.500 (12.70)	.450 (11.43)	.400 (10.16)

1/ Complete PIN shall include additional symbols to indicate capacitance tolerance (K, J, G, or F) and failure rate level (M, P, R, or S), as applicable.
2/ Metric equivalents are in parentheses.
3/ T, H, and L are bare case dimensions (see figure 1). When specified on the purchase order, insulating covering for both "L" and "T" dimensions shall not exceed .025 inch (0.64 mm), nor shall the insulating covering for "H" dimension exceed .040 inch (1.02 mm).

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet) 125 percent of dc rated voltage applied. See MIL-PRF-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's), 50 percent of dc rated voltage applied.

Salt atmosphere (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: Five percent.

Immersion: Method 104 of MIL-STD-202, condition B.

DWV: Terminal to terminal: 200 percent of rated voltage.

IR: Terminal to terminal: Not less than 50 percent of value specified for insulation resistance.

Capacitance: Within ±1 percent of initial value.

DF: Not more than .3 percent.

Solderability: Method 208 of MIL-STD-202, the following details shall apply:

Number of terminations of each capacitor to be tested: Two.

Depth of immersion in flux and solder: both terminals shall be immersed to within .050 inch (1.27 mm) of the capacitor body.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I, 50 percent of dc rated voltage applied. No arcing or other evidence of electrical breakdown.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260°C ±5°C for 10 ±1 seconds).

IR: See initial value.

ΔCapacitance: Within ±0.25 percent of initial value.

DF: See initial value.

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Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, Capacitance, and DF: Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and C.

Condition A: Applied force 5 pounds.
Condition C: Applied force 5 pounds.

Low temperature and capacitance change with temperature:

Low temperature: In accordance with MIL-PRF-39022.

Capacitance change with temperature (maximum):

At -55°C: 0 to -1.7 percent.

At +25°C (retrace): ±0.25 percent.

At +125°C: +1.2 percent to -0.7 percent.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202, condition B.

ΔCapacitance: Maximum of ±0.25 percent.

Life (at +100°C): Method 108 of MIL-STD-202.

Accelerated conditions: 140 percent of dc rated voltage for 2,000 +72, -0 hours.

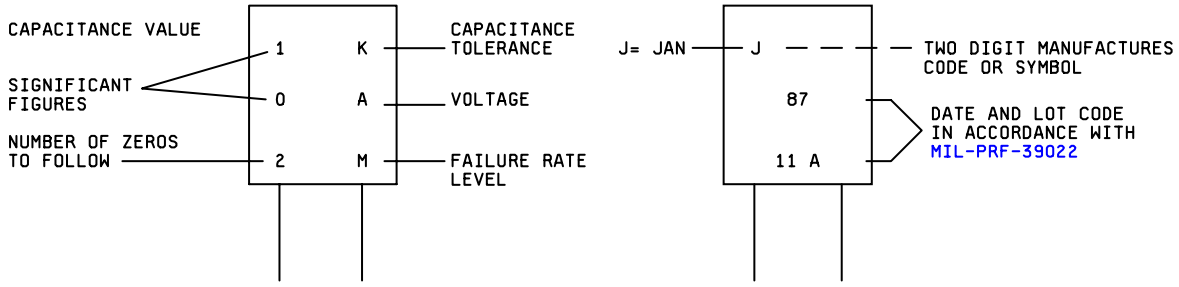
Rated condition: 100 percent of dc rated voltage for 10,000 +96, -0 hours.

DF: (At +100°C) between 24 and 48 hours of test - see initial requirement.
IR: Terminal to terminal not less than 33.3 percent of initial requirement.

The current to each capacitor will be limited to 1 ampere maximum by a series resistor.

Marking: Marking shall be in accordance with MIL-PRF-39022, except the minimum marking as shown in the following example will be permitted. Full marking shall be included on the package.

EXAMPLE OF MINIMUM MARKING



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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents: In addition to MIL-PRF-39022, this document also references MIL-STD-202 and MIL-STD-810.

Custodians:
Army - CR
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2008-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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